Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/31/2016
PI Name:	Oates, Chris Ph.D.		
Project Title:	Development/Evaluation of a 10^-17 Yb Optical Clock for Space Applications		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/31/2018 per PI (Ed., 3/7/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Diddams, Scott (National Institute of Standards & Technology) Ludlow, Andrew (National Institute of Standards & Technology)		
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Task Description:	The primary goal of the European Space Agency's (ESA) Space Optical Clock (SOC) program is to develop a high-performance optical frequency standard (clock) for operation in space with 1 x 10^-17 frequency uncertainty. National Institute of Standards and Technology (NIST) proposes to leverage its existing research on world-leading Yb lattice optical clocks to: (1) demonstrate a laboratory-based Yb lattice optical clock with frequency uncertainty of 1 x 10^-17 or better; (2) stabilize a space-friendly fiber frequency comb to the Yb clock signal; and (3) develop a preliminary design for space-qualification by providing clock subsystem specifications. The proposed program will provide ESA with detailed information to make the most informed decisions on use of one or more optical frequency standards for the flight portion of the SOC program. Currently, ESA has identified the Sr and Yb lattice optical clocks as the most promising technologies. A Yb lattice clock has significant potential advantages over Sr in that Yb clocks can operate with solid state lasers much more likely to be practical in space operations. Several European groups are presently evaluating Sr's space prospects and Sr has thus received significant attention. For Yb, however, the most mature and high performance lattice clock resides in our lab at NIST. This Yb clock currently performs at least as well as the world's best Sr clocks, approximately 10^-16 frequency uncertainty in all cases, and continues to rapidly improve. Thus, we are uniquely qualified to play a vital role in the SOC project. For a modest investment, the proposed research program will ensure ESA has the best information and options to select the optical clock technology to meet SOC program goals, ensuring fully informed decisions affecting a much larger financial and scientific investment in the SOC program. Furthermore, by demonstrating a 10^-17 level Yb optical lattice clock, our proposed effort will achieve significant milestones currently outlined in the SOC program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NIST SOC program (Yb lattice clock) also directly supports the major benefits of the overall Atomic Clock Ensemble in Space (ACES) program by developing a new high performance NIST optical frequency standard that will directly contribute to NIST's roles in the broader ACES program: 1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. The NIST Yb lattice clock represents a dramatic improvement in stability and accuracy compared to cesium microwave primary frequency standards. 2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory. 3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, search for possible Lorentz transformation variations, and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosomology.
	Progress in the most recent reporting period (April 2015 through March 2016) has been substantial, with several tasks ahead of schedule, and we are on target to complete the most challenging task, that is, an exhaustive clock evaluation with a total fractional frequency uncertainty below 10^-17. The major advances for this period include: 1. Reduction of key clock systematic shifts to the 10^-18 level: blackbody/electric/magnetic field effects and lattice Stark shifts Reaching the stated ambitious goal of a total clock fractional frequency uncertainty below 10^-17 requires careful evaluation of all relevant systematic effects near the 10^-18 level (so that the combination of all the individual uncertainties remains below 10^-17). For lattice clocks, shifts due to three particular field effects (background blackbody radiation, stray electric/magnetic fields, and lattice trapping light) are the most formidable to be minimized and measured. In last year's report we described how we constructed a copper shield to surround the atoms with a knowable blackbody environment. We were then able to evaluate and quantify the blackbody shift contribution with a 10^-18 uncertainty, thereby removing this shift from the "to-do" list. This year we were able to remove the electric/magnetic fields and the lattice light shifts from the list as well. DC Electric/Magnetic Field Shifts: The key to minimizing electric field effects was to design the blackbody shield so that it behaves also like a Faraday cage, which shields the atoms from stray dc electric fields. In fact, the cage works so well that it was hard to generate sufficiently large (and reversible) artificial electric fields to verify the shielding performance. In the end we were able to set a limit of 8 x 10^-21 for the fractional frequency shifts due to stray fields in the vertical direction and of limits of 3.5 x 10^-19 and 5 x 10^-19 for two orthogonal horizontal directions, thereby verifying the shielding performance of the enclosure. In a similar way we have evaluated the effect of stray magnetic fields on the clock frequency. The transition we use has a weak linear sensitivity (~½ x 400 Hz/G, due only to the nuclear g factor) and a still weaker quadratic sensitivity (~0.1 Hz/G2). We remove the linear dependence directly by locking to the average of two transitions with equal and opposite linear Zeeman shifts. This process also allows us to know accurately the size of the field, which is set by a weak bias field (~1 G) and any residual stray fields that are not cancelled by our three-dimensional set of bias coils. Through a series of measurements with varying (and reversible) applied magnetic fields, we were able to determine the coefficient for the quadratic shift (-0.06100(13) Hz/G2). We could then calculate a total fractional frequency uncertainty for magnetic field related effects (including vector light shifts), which came in < 5 x 10^-19.

Task Progress:	Lattice Light Shifts: The most significant shift for lattice clocks results from the lattice light that is used to confine the atoms and suppress the Doppler shift. The solution is to choose a clock transition and a lattice wavelength so that the linear (in intensity) light shift term is common mode to both the ground and excited states for the clock transition, thereby leaving the clock transition effectively unperturbed. This “magic” wavelength lattice approach works amazingly well, but for clocks with a fractional uncertainty approaching 10^-18, it is necessary to consider in detail higher-order effects that lead to deviations from this model, some of which have exotic, square-root dependencies. During this period, we found that we could reach the sub-10^-18 level through several a series of improvements to the apparatus, and finally, the most detailed set of measurements to date of the Stark shifts for a lattice clock. The key improvements were to work with shallower lattices (less than 50 atomic recoils) in a vertical orientation and to filter the lattice light spectrum with extremely narrow bandpass filters (30 GHz) to minimize shifts due to asymmetric amplified spontaneous emission spectra. The crucial piece to our high precision measurements of the shift was the discovery that when we actually measure these effects by varying the lattice intensity, the temperature of the atomic sample also changes and (fortuitously) in such a way that the dependencies of these more exotic effects revert to either linear or quadratic. As a result, we were able to find the magic wavelength (linear zero point) with unprecedented accuracy, and determine the quadratic coefficient as well. Finally, we found it better to operate slightly detuned from the magic wavelength, at a point where the linear and quadratic terms have opposite signs, thereby yielding a flat region around our operating lattice intensity. 2. Demonstrated improved Yb clock stability due to a new clock laser system In the past year we have begun to bear much fruit from the new laser/cavity system that we installed the previous year. The new system is based on frequency doubling of an infrared semiconductor laser and an improved reference cavity with sub-kHz linewidth. Measurements with Yb clock transition revealed that the new laser system enabled sub-Hz linewidths and a clock stability nearly a factor-of-three improvement over our results published nearly two years ago in Science. We sent light from this improved clock source to both Yb clocks and performed a series of clock comparisons with the cycles for the two clocks either synchronized (i.e., we excite the clock transition simultaneously in the two systems) or asynchronized (i.e., we interleave the excitation periods between the two clocks). Asynchronous measurements reveal the true clock instability, which we found to be ~1.5 x 10^-16 t^-1/2, the lowest fractional frequency clock instability measured to date. Synchronized measurements yield still lower results (~7 x 10^-17 t^-1/2, averaging down to 10^-18 in only 4000 s), since the laser frequency noise is common mode. Given the absence of this noise source, synchronized results are not a true measure of single clock instability, but this mode can be used to improve clock comparisons and will enable more efficient investigations of systematic effects. We also combined the two atomic apparatus into a single clock, operated in such a way that spectroscopic interrogation of each atomic ensemble is interleaved in time. Doing so creates a single, composite clock with zero-dead-time, thereby greatly suppressing frequency noise. We note that this type of zero-dead-time operation with two atomic samples may be especially meaningful for operation of the lattice clock outside the laboratory environment, such as in space conditions, because we can relax laser cavity requirements that are hard to fulfill in noisy environments. These results represent an important step forward in state-of-the-art clock operation, enabling very high measurement precision in a relatively short averaging time. 3. Reduction of the temperature of the lattice-trapped atoms via sideband cooling This year we also had a bonus experimental achievement with the Yb clock apparatus in that we were able to significantly reduce longitudinal temperature of the atoms trapped in the optical lattice, which is particularly valuable for clocks based on Yb. While Yb has several distinct advantages (e.g., smaller blackbody shift, larger mass, easier second stage laser cooling) over its main competitor (Sr) in the optical lattice clock game, its principal disadvantage has been higher atom temperatures. Higher atom temperatures have several deleterious effects: they reduce our signal contrast and they require deeper lattices to confine the atoms, thereby increasing the lattice light shift. Borrowing a technique originally developed for ions here at NIST and since demonstrated in other clock systems, we have implemented a cooling approach, termed “quenched sideband cooling,” which has enabled us to reduce the atom temperature from ~6 μK to below 1 μK with less than 10 ms of cooling. We find that we get colder temperatures (some spectra indicate sub-microkelvin temperatures) at a lower lattice depth (equivalent to ~25 atomic recoils), which offers the additional benefit of reduced lattice light shifts. We can now envision that with the addition of only 10 ms of atom preparation time, we can work with much cooler atomic samples that offer negligible tunneling effects and operate with a lattice depth that yields a total fractional frequency shift for the clock transition of only 2 x 10^-18 (with a corresponding uncertainty of less than 5 x 10^-19).
Bibliography Type:	Description: (Last Updated: 04/05/2023)
Articles in Peer-reviewed Journals	Ludlow AD, Boyd MM, Ye J, Peik E, Schmidt PO. "Optical atomic clocks." Review of Modern Physics. 2015 Apr-Jun;87(2):637-700. Review. http://dx.doi.org/10.1103/RevModPhys.87.637 , Apr-2015
Articles in Peer-reviewed Journals	Ye J, Ludlow AD. "Progress on the optical lattice clock." Comptes Rendus Physique. 2015 Jun;16(5):499-505. http://dx.doi.org/10.1016/j.crhy.2015.03.008 , Jun-2015
Awards	Ludlow AD. "Presidential Early Career Award for Scientists and Engineers (PECASE), February 2016." Feb-2016